

Conformal Perceptual Shielding for Adaptive Bitrate Streaming

Saeed Zarbi^a, Leili Farzinvash^{a,*}, Pedram Salehpour^a

^aDepartment of Computer Engineering, Faculty of Electrical and Computer Engineering, University of Tabriz, Tabriz, Iran

Abstract

Client-side adaptive bitrate (ABR) controllers decide chunk by chunk. A single request can stall playback when throughput drops, or waste bandwidth when the top rungs of the bitrate ladder are perceptually saturated. Standard runtime repairs address neither problem: they ignore perceptual saturation and offer no coverage guarantee. We propose the Certified Perceptual Shield (CPS), a model-agnostic projection that wraps heuristic, model-based, or learned ABR policies. First, knee-based bandwidth banking lowers even a feasible proposal to the smallest rung whose Video Multimethod Assessment Fusion (VMAF) score lies within a perceptual budget of the proposal. Second, an online split-conformal lower bound on throughput provides distribution-free coverage under residual exchangeability. Third, a feasibility check prefers rungs whose download time under that bound fits the buffer margin. If none does, or if the buffer is below a critical level, CPS still emits the lowest rung. The stall statement is then a per-chunk probabilistic certificate only when the **executed rung** meets the test and the buffer is at **least 0.1 s**, not a zero-stall promise. We evaluate CPS on 204 paired synthetic 5G episodes over twelve titles with diverse rate–quality ladders. Compared with a safety-only shield, banking reduces delivered bitrate by 3.8–4.5% and rebuffering by 8.9–9.3% for four deterministic controllers. Mean VMAF stays within one point (equivalence test). Empirical coverage is 0.919 against a 0.90 target, and the first 20 of 48 chunks in each episode use a fixed fallback scale. As a supporting mechanism study, risk-aware predictive banking (CPS-P) reaches 14.6% bitrate and 12.6% rebuffering savings versus safety at the default budget; a larger stall = 2 cuts rebuffering further. **On real fixed and mobile broadband traces, banking gains fall to about 0.9%. A feasibility projection removes about 98.7% of stall time versus unshielded greedy, which already stalls for most of those sessions. Banking therefore pays off where ladders saturate. The broadband stall cut is a feasibility effect, not evidence that the conformal bound transfers.**

Keywords: adaptive bitrate streaming, runtime shielding, VMAF, conformal prediction, certified safety, video QoE

1. Introduction

Client-side adaptive bitrate (ABR) policies target average quality of experience (QoE). Several controller families pursue this target. Buffer-based adaptation (BBA) [1] and BOLA [2] use buffer or Lyapunov rules. RobustMPC [3] **plans** over a short throughput horizon. Bitrate-greedy selection always requests the top rung, and proximal policy optimization (PPO) [4, 5] learns a policy from traces. Session averages, however, can obscure transient risks associated with individual chunks. A single segment can still stall under volatile sub-6 GHz and mmWave 5G throughput [6–8]. A comfortable buffer creates the opposite problem: wasted bandwidth. Commercial rate ladders often saturate in Video Multimethod Assessment Fusion (VMAF) [9]: further bitrate increases at the top often provide only marginal perceptual gains. Requesting those rungs when the buffer is healthy consumes bandwidth that could cushion the next dip.

A runtime shield sits between the base policy and the player [10–12]. It checks each proposed representation before download and replaces an unsafe proposal. The standard repair decrements the **representation index** until a buffer-feasibility test passes. That repair is content-blind, because it ignores

VMAF saturation. Prior shields often apply a fixed throughput scale without a stated coverage guarantee. **CPS** uses the same kind of fallback scale only during a short warm-up ($W_{\text{calib}}=20$ chunks), then switches to an online split-conformal lower bound with finite-sample coverage. Deployed ladders carry exploitable structure—rate–quality saturation near the top. That structure supports perceptually negligible bitrate reductions while freeing additional buffer capacity.

We introduce a *Certified Perceptual Shield* (CPS) that exploits this structure while stating a formal coverage guarantee:

1. **VMAF-knee bandwidth banking.** Even a feasible action is lowered to the smallest rung whose VMAF lies within a budget ϵ in VMAF points, of the proposal. The downshift is perceptually bounded, and reclaimed bytes bank as buffer for later chunks.
2. **Conformal throughput lower bound.** Download-time tests use an online split-conformal lower bound on throughput. Under residual exchangeability, the bound covers the next chunk with probability at least $1-\alpha$, where α is the tolerated miss rate. Calibration uses recent residual ratios [13, 14].
3. **Certified feasibility.** After banking, CPS prefers a rung whose download time under the lower bound fits the buffer margin. If no such rung exists, or if $B_k \leq B_{\text{crit}}$, it still emits the lowest rung. When the executed rung meets the test

*Corresponding author.

Email address: l.farzinvash@tabrizu.ac.ir (Leili Farzinvash)

and $B_k \geq 0.1$ s, the result is a per-chunk probabilistic stall certificate at level $1-\alpha$, rather than a deterministic no-stall guarantee.

On a monotone ladder, VMAF never decreases with the rung index. A shield that only ranks rungs by VMAF then picks the highest feasible rung, **so a perceptual-loss budget cannot change its choice**. The same projection attaches to bitrate-greedy, BBA, and PPO controllers alike.

The main contributions are:

- **Certified Perceptual Shield.** A model-agnostic runtime projection that combines VMAF-knee bandwidth banking, a conformal throughput lower bound with finite-sample coverage under residual exchangeability, and certified buffer feasibility.
- **Isolated banking evaluation.** Paired **RAW, SAFETY**, and **CERTIFIED** rollouts on greedy, BBA, BOLA, RobustMPC, and PPO hosts. We report Wilcoxon tests on bitrate and rebuffering, and **two one-sided tests** (TOST) of VMAF within $\pm\epsilon$ [15].
- **Optional predictive banking (supporting).** A buffer- and look-ahead-conditioned perceptual budget (CPS-P) that widens to ϵ_{risk} under risk. Section 5.3 compares it with static ϵ sweeps; **a larger static budget** can cut rebuffering further.
- **Regime characterization.** Banking gains concentrate where ladders saturate at the top and hosts request those rungs (primary 5G suite). Real-trace broadband stress is dominated by safety projection.
- **Policy-shield co-design (supporting).** An optional single-seed PPO study with CPS in the training loop. We report the QoE weight w^* at which this co-trained policy crosses **shield-at-eval-only** operation, but we do not treat the arm as a headline claim.

To separate banking from safety, we run three paired arms on identical episode seeds: RAW (unshielded), SAFETY (conformal feasibility only), and CERTIFIED (feasibility plus banking). The banking effect is the CERTIFIED minus SAFETY contrast. The safety effect compares SAFETY or CERTIFIED against RAW. We run 204 paired 5G episodes over twelve titles, with a perceptual budget $\epsilon=1.0$ and a tolerated miss rate $\alpha=0.10$. Certified banking reduces mean bitrate and rebuffering on greedy, BBA, BOLA, and RobustMPC (Table 5). For greedy, banking reduces bitrate by 4.5% and rebuffering by 9.3%. Mean VMAF stays within $\pm\epsilon$ by TOST, and empirical coverage is 0.919 against a 0.90 target. Optional CPS-P and static ϵ sweeps appear in Section 5.3. On real fixed and mobile broadband traces from the FCC Measuring Broadband America program and the Norway/Nornet logs [16, 17], banking gains narrow to about 0.9% bitrate reduction (certified vs. safety). Conformal feasibility still reduces stalls by about 98.7% vs. raw greedy on those traces. That host already stalls for most of the session, so the cut is a

feasibility effect. These results indicate where perceptual banking is effective. Optional PPO co-training with CPS remains a supporting, single-seed study.

The remainder of this paper is organized as follows. Section 2 reviews related work, and Section 3 formalizes the streaming model. Section 4 presents CPS and optional co-training. Section 5 reports the evaluation, Section 6 discusses limitations, and Section 7 concludes.

2. Related work

Three areas bear on client-side representation selection: model-based ABR, learned ABR, and constrained control with runtime shielding. Our focus is the control policy and its runtime action projection; throughput prediction lies outside this scope. Predictors such as CS2P [18] play a complementary role: they feed inputs to model-based or learned controllers [3, 4]. ABR surveys [8, 19, 20] and 5G/mmWave measurements [6, 21] agree on one empirical lesson: control must remain robust when throughput fluctuates sharply.

2.1. Heuristic and model-based ABR

MPEG Dynamic Adaptive Streaming over HTTP (DASH) separates packaging from transport [22]. Clients execute local adaptation policies based on ladder metadata and observed download conditions. Buffer-threshold heuristics such as BBA [1] remain simple and interpretable. BOLA [2], by contrast, derived its decisions from a Lyapunov-based utility formulation. RobustMPC [3] planned ahead. It selected representations by solving a short-horizon model predictive control (MPC) problem under estimated future throughput. The payoff is interpretable thresholds and horizons. The cost is fragility. These methods depend on manual calibration and on throughput predictions that vary across networks and devices [6, 7]. Moreover, VMAF scores saturate on the upper rungs of a ladder, so those rungs add little perceptual quality [9]. Conventional feasibility rules rarely exploit this saturation to reserve throughput headroom in advance. Other related efforts target different stages of the delivery pipeline. SODA [23] pursued smoothness-aware online control. Edge assistance [24] and Oboe-style automatic tuning [25] addressed other parts of the pipeline. Recent client-side designs also treat throughput uncertainty explicitly rather than relying on a single point forecast. Conformal-ABR [26] wraps a throughput predictor with split-conformal intervals. It selects a representation under an explicit stall-risk budget. This yields distribution-free control of prediction error when the traces are exchangeable. SafeSABR [27] targets volatile satellite broadband with risk-calibrated reinforcement learning. A runtime safety auditor then checks each policy proposal against conservative capacity bounds before download. Kairos [28] instead strengthens MPC-based ABR with streaming-aware prediction and buffer-conditioned uncertainty, improving robustness without a conformal certificate. These lines of work address complementary failure modes: risk-aware control, runtime auditing, and robust planning. None of them combines all three of our ingredients. These are a model-agnostic post-shield that overwrites

an unsafe proposal, a stated conformal throughput lower bound, and VMAF-knee bandwidth banking on saturated ladders. CPS fills that gap by separating an optional performance policy from a deterministic projection that enforces both certified feasibility and perceptual banking at every chunk.

2.2. Deep reinforcement learning for ABR

Pensieve [4] established the template: a trace-driven DRL formulation for ABR. D-DASH [29] and more recent PPO-based controllers [30] followed this line. Their neural policies combine throughput histories, buffer state, and manifest features [31]. Comyco [32] trained by imitation against a VMAF-based QoE objective and showed that perceptual signals helped inside the learned policy. Loki [33] combined learned features with a rule-based model and reduced long-tail failures in production. Its rule module limits extreme decisions, but it provides no formal feasibility certificate. Fugu [34] deployed learned control in live networks. It exposed the distribution shift between simulation and deployment. That shift motivates runtime safeguards beyond expected-return training. Taken together, these systems motivate constrained learning and decision-time enforcement. An expected-return objective alone encodes no explicit per-episode stall or smoothness budget [12, 35]. Explicit constraints do not fully close this gap. A controller trained under a constrained Markov decision process (CMDP) retains guarantees only over the sampled trajectory distribution. An unseen blockage pattern can still induce an unsafe proposal at a single chunk boundary. This gap argues for splitting the stack into a learned performance layer and a runtime enforcement layer, because additional training cannot substitute for runtime validation.

2.3. Safe/constrained RL, shields, and the content-blindness gap

CMDPs and Lagrangian primal–dual methods optimize expected return under explicit constraints [36–38]. Safe-RL surveys catalogued related families: shielding, Lyapunov, and risk-sensitive schemes [12, 39]. COREL [35] is closest to our training formulation. It cast ABR over commercial 5G/mmWave traces as a CMDP with a stall-time constraint. The CMDP is solved by a primal–dual actor–critic. That constraint, however, operated within training. It is not a standalone runtime projection that logs each intervention at decision time. Runtime shielding takes the opposite approach: it repairs a proposed action before execution and can log each intervention [10]. Surveys split the family into *pre-shielding*, which prunes the action set, and *post-shielding*, which overwrites an unsafe proposal [11]. By that taxonomy, CPS is a post-shield. Its safety set is drawn by a conformal feasibility test rather than a temporal-logic model.

This distinction shapes our design: we separate an optional learned performance layer from a deterministic runtime CPS. Learned policies such as Lagrangian PPO can pursue expected stall and smoothness budgets across training rollouts [35, 37]. They cannot certify per-chunk feasibility under conformal throughput bounds. CPS targets exactly this gap. Before every download it (i) banks bandwidth on VMAF-saturated ladder rungs within a perceptual budget ε , measured in VMAF points, and (ii) projects onto actions feasible under a distribution-free

Table 1: Positioning of client-side ABR methods by training constraints, deterministic runtime projection, conformal throughput bounds, and VMAF-knee banking.

Work	Year	Paradigm	Constr.	Guard	Conf.	Bank
BBA [1]	2014	Heuristic	N	N	N	N
BOLA [2]	2018	Lyapunov util.	Imp.	N	N	N
RobustMPC [3]	2015	MPC/lookahead	N	N	N	N
Pensieve [4]	2017	DRL	N	N	N	N
Comyco [32]	2019	Quality-aware IL	N	N	N	N
Loki [33]	2021	Rule–RL fusion	N	$\approx$	N	N
COREL [35]	2023	Constr. DRL	Y	N	N	N
SODA [23]	2024	Online control	N	N	N	N
Conformal-ABR [26]	2026	Risk-controlled ABR	N	Y	Y	N
SafeSABR [27]	2026	Risk-cal. RL + auditor	Y	Y	$\approx$	N
Kairos [28]	2025	MPC + uncertainty	N	N	N	N
Ours (CPS)	—	Shield ($\pm$ RL)	$\approx$	Y	Y	Y

Constr.: explicit stall budget in training (Y), implicit (Imp.), optional/partial ($\approx$), or none (N). **Guard:** auditable action projection before download [10].

Conf.: conformal throughput lower bound with stated coverage [13]. **Bank:** VMAF-knee bandwidth banking within a perceptual budget ε .

throughput lower bound with coverage $1-\alpha$, where α is the tolerated miss rate [10, 13, 14]. Its use of perceptual data also differs from Comyco, where VMAF enters only the learned objective [32]. CPS reads the ladder at decision time to bank bandwidth proactively, rather than only repairing the representation index in an emergency. Table 1 compares representative client-side methods on training constraints, runtime guard, conformal bounds, and VMAF-knee banking. A Y under Conf. marks a stated conformal coverage guarantee; a Y under Bank. marks proactive perceptual banking.

3. System model and problem formulation

This section formalizes client-side ABR at the chunk timescale. We assume a single client, exogenous trace throughput C_k^{tr} , and deterministic buffer updates given $(s_k, a_k^{\text{exec}}, C_k^{\text{tr}})$; cross-layer scheduling and transport adaptation are out of scope. A base policy proposes a ladder index; CPS applies a deterministic projection before download; the player then updates the buffer under realized throughput. The subsections below define the architecture, the state–action–reward tuple, the ladder saturation that banking exploits, and the download and buffer dynamics. CPS internals—the conformal bound, safety set, and knee rule—are deferred to Section 4 so $\underline{C}_k$ is defined before use. Table 2 collects the symbols used throughout.

3.1. Streaming architecture and control loop

We work in the standard Hypertext Transfer Protocol (HTTP) adaptive-streaming abstraction [3, 4, 22]. A video origin or content delivery network (CDN) publishes L encoded representations. We index them by $\mathcal{B} = \{0, \dots, L-1\}$; the Dynamic Adaptive Streaming over HTTP (DASH) or HTTP Live Streaming (HLS) manifest carries the matching per-chunk sizes and perceptual metadata. At decision step k , the client fetches chunk

Table 2: Summary of notation.

Symbol	Meaning
k, T_c, K_e	chunk index, chunk duration, chunks per episode
$L, \mathcal{B}=\{0, \dots, L-1\}$	number of ladder rungs, rung index set
$a_k^{\text{raw}}, a_k^{\text{exec}}, \iota_k$	proposed rung, executed rung, intervention bit
B_k, m, B_{crit}	buffer level, safety margin, critical buffer
$V_k(j), \sigma_k(j)$	VMAF and size (bits) of rung j at chunk k
$\varepsilon, \varepsilon_{\text{eff}}, \varepsilon_{\text{risk}}$	perceptual budget, effective/risk-mode budget
$C_k^{\text{tr}}, \hat{C}_k, C_k^{\text{act}}, \underline{C}_k$	emulator trace rate; conformal predictor; realized; lower bound
$W_{\text{calib}}, \rho_{\text{fb}}$	residual warm-up length, fallback scale before calibration
$\alpha, 1-\alpha$	conformal miscoverage, target coverage
$r=C^{\text{act}}/\hat{C}, W, \kappa$	residual ratio, residual window, predictor horizon
$d_k(j), H, \varepsilon_0$	download time of rung j , look-ahead, numerical guard
μ, β	logged-QoE smoothness and stall weights ((2))
w, w^*	QoE stall weight (sweep), mean-QoE crossover weight

k while holding a playout buffer B_k . The effective trace throughput C_k^{tr} is exogenous to the controller: the ABR client reacts to measured delivery rates, not to radio-scheduler internals. This matches mobile access traces where blockage and mobility shift capacity abruptly [6, 7].

A base policy π (heuristic or learned) maps an augmented state s_k —buffer occupancy, throughput history, manifest lookahead, and the VMAF ladder (defined below)—to a proposed action $a_k^{\text{raw}} \in \mathcal{B}$. Before chunk k is downloaded, the Certified Perceptual Shield $\mathcal{S}$ applies the deterministic projection

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

The projection returns the executed representation index a_k^{exec} and an intervention bit ι_k , which indicates whether the projection altered the proposal. Only a_k^{exec} feeds the chunk download and buffer dynamics (4); Fig. 1 traces the $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path.

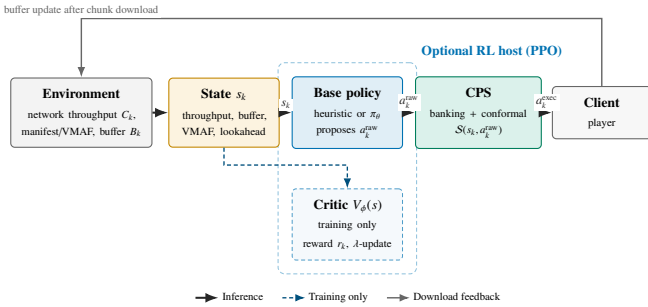


Figure 1: Control loop. The base policy proposes a_k^{raw} ; CPS banks on the VMAF knee and projects the proposal to a_k^{exec} using the conformal lower bound $\underline{C}_k$ (Section 4). Training-only dashed paths (critic, dual variables) are optional for RL hosts.

Decisions advance on the chunk clock, one step per playback duration T_c (4 s in our emulator) [3, 4].

The two layers serve distinct roles. Optional learned-host training (Section 3.5) handles stall and smoothness budgets in expectation over rollouts [37, 38]. However, it does not guarantee per-step stall avoidance from throughput uncertainty alone. CPS supplies a separate layer: given a conformal lower bound $\underline{C}_k$ (Section 4), it deterministically projects a_k^{raw} to a feasible rung; marginal coverage of $C_k^{\text{act}} \geq \underline{C}_k$ is probabilistic at rate $1-\alpha$ [10, 13].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . The observation holds buffer occupancy, past throughput samples, and the previous decision. It also includes manifest lookahead: upcoming chunk sizes across the representation ladder, plus spatial complexity (SI) and temporal complexity (TI) indices [3, 4, 22] (higher SI/TI indicate richer spatial or temporal structure in the chunk). The VMAF ladder $\{V_k(j)\}_{j \in \mathcal{B}}$ completes the observation, where $V_k(j)$ is the perceptual score for chunk k at representation j [9]. This last component is essential: without $\{V_k(j)\}$, neither a learned host nor CPS can detect the content-dependent ladder saturation that bandwidth banking exploits.

Nonlinear, content-dependent fidelity. Saturation arises because quality does not scale linearly with rate: for each chunk k , the map $j \mapsto V_k(j)$ is not affine in the discrete index j . This map also varies across chunks. Different group-of-pictures (GOP) and texture structures yield different marginal VMAF gains between adjacent rungs. Therefore, the bitrate-to-VMAF relationship is both nonlinear and content-dependent. Consequently, adjacent rungs may differ by a few VMAF points or by more than ten. Quality often saturates at the upper rungs of the ladder—precisely the structure that CPS banking exploits [9]. Perceptually aware, JND-based per-title encoding aims to remove this saturation at authoring time [40]. In contrast, CPS exploits any saturation that remains in a deployed ladder at decision time.

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed representation index for chunk k ; the environment executes only a_k^{exec} .

QoE surrogate. Keeping to RobustMPC/Pensieve reporting conventions [3, 4], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \mu \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ is the absolute VMAF change between adjacent chunks and τ_k^{stall} is the stall duration at step k . The first chunk takes a zero smoothness term. Our shared emulator fixes $\beta=4.3$ per stall second and $\mu=1.0$ per absolute VMAF-point change. The symbol α remains reserved for the conformal miscoverage throughout. The rebuffering weight is a Pensieve-style penalty, rescaled onto the VMAF fidelity axis. Session QoE sums over chunks: $\sum_{k \in \mathcal{K}_c} \text{QoE}_k^{(\text{rep})}$ (Section 5). For a learned host, optional RL training substitutes a Lagrangian CMDP objective over rollouts (Appendix A) [36, 37].

3.3. Ladder saturation

The evaluated system reports one VMAF value per representation and video (pooled per-frame scores), so shield decisions operate on a per-title session-mean ladder that does not vary with chunk index k . Table 3 summarizes session-mean ladders for 12 publicly available reference titles (five Blender animations and seven natural-motion Xiph DERF clips, including Ducks Take Off); Fig. 2 illustrates per-chunk spread on a four-title subset with frame-level VMAF logs (Big Buck Bunny, Tears of Steel, Sintel, and CrowdRun). CrowdRun is not among the twelve titles in Table 3. Across the twelve-title suite, 67% exhibits a non-monotone session-mean ladder. Mid-ladder inversions concentrate at the 750→1200 kbps step; the top three rungs (1850,

Table 3: Session-mean representation ladder across twelve evaluation titles.

Video	Held-out	SI	TI	Top gain	Span	Mono.
Big Buck Bunny	—	60.6	47.1	+0.03	30.6	Yes
Tears of Steel	—	41.0	31.8	+4.09	26.1	No
Sintel	✓	50.4	14.7	+1.30	22.9	Yes
Elephants Dream	—	51.8	27.0	+7.87	31.2	No
Ducks Take Off	—	94.9	16.6	+11.59	36.6	No
Cosmos Laundromat	✓	42.6	38.3	+5.23	34.6	No
Park Joy	—	119.7	34.8	+14.31	46.5	No
Old Town Cross	—	61.2	12.6	+6.85	49.5	Yes
Rush Hour	—	27.7	11.7	+4.00	29.8	No
Into Tree	—	47.4	14.4	+9.12	36.1	No
Sunflower	—	38.0	16.5	+1.42	24.9	No
Kristen & Sara	✓	85.8	7.0	+1.53	37.6	Yes

Top gain is VMAF(6000)–VMAF(2850) in kbps; Span is VMAF(6000)–VMAF(300). Mono. marks a non-decreasing session-mean ladder. Held-out titles (✓) are excluded from PPO training. SI/TI are spatial/temporal complexity indices.

2850, 6000 kbps) remain nondecreasing on every title. The mean top-rung gain (6000 vs. 2850 kbps) is only 5.61 VMAF points. This pattern reflects strong saturation on animation titles (e.g. Big Buck Bunny at +0.03) but steeper climbs on natural content (e.g. Park Joy at +14.3). Two properties make banking worthwhile. First, many adjacent pairs differ by at most one VMAF point on saturated titles, so a single-step downgrade costs almost nothing perceptually. Second, the top rungs saturate frequently, allowing the knee rule (7) to bank bytes with no material quality loss. On the four titles with a nondecreasing session-mean ladder (Table 3), max-VMAF and max-index feasible choices coincide (Proposition 2). The same equivalence holds for the upper three rungs on every title; greedy banking operates mainly there. CPS instead picks the smallest rung within the perceptual budget, preserving near-lossless downshifts under saturation. Chunk-level inversions appear in Fig. 2 for the logged subset. However, banking does not depend on them. Section 5.5 shows that the knee rule remains effective. It banks more when the shield observes the raw per-chunk ladder.

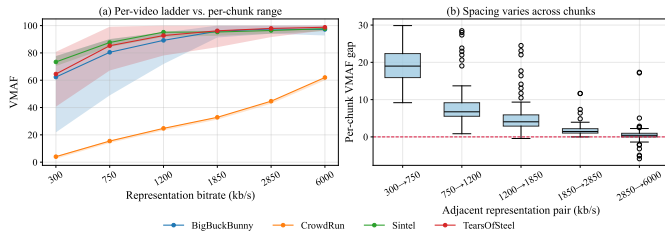


Figure 2: Representation ladder from per-frame VMAF on four titles with frame logs (Big Buck Bunny, Tears of Steel, Sintel, CrowdRun; 173 chunks). CrowdRun is absent from the twelve-title session-mean table. The banking ablation in Section 5.5 uses 113 chunks from a different per-chunk export of the same four-title family. (a) Per-video pooled ladder (markers) against the per-chunk range (shaded). (b) Adjacent-rung VMAF gaps; the dashed line marks a zero gap.

3.4. Chunk-timescale dynamics

3.4.1. Throughput and latency

Let $C_k^{\text{tr}} > 0$ be the trace-derived effective throughput, in kb/s, for chunk k . Section 4 defines a separate predictor $\hat{C}_k$ (harmonic mean of recent realizations) that enters $\underline{C}_k$. C_k^{act} is the realized

throughput of the executed download. Every method sees the same throughput trace in paired evaluations. The choice of a_k^{exec} sets the manifest payload size $\sigma_k(a_k^{\text{exec}})$ in bits. Write $\tilde{C}_k = \max\{C_k^{\text{tr}}, \varepsilon_0\}$ for a small constant $\varepsilon_0 > 0$ (the same numerical guard as in (5)), and $d_k(j) = \sigma_k(j)/(10^3 \tilde{C}_k)$ for the nominal download duration of rung j . The executed chunk uses $d_k \equiv d_k(a_k^{\text{exec}})$ [3]. Feasibility timing under $\underline{C}_k$ is defined in Section 4.

3.4.2. Buffer and stall

Let $B_k \geq 0$ be the buffered playback time when the download of chunk k begins [3, 4]. Rebuffering (stall) and the buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Randomness across episodes has a single source: trace and initial-condition sampling [16, 17, 19]. Given $(s_k, a_k^{\text{exec}}, C_k^{\text{tr}})$, each transition is deterministic.

3.5. Optional Lagrangian training for RL hosts

When the base policy is learned, we train PPO with a Lagrangian-relaxed constrained MDP (CMDP) objective that penalizes rebuffering and smoothness alongside fidelity [36–38]. The dual variables adapt as training proceeds; Appendix A details the optional buffer-deviation and Lyapunov stabilizers [41]. None of this certifies per-chunk feasibility, which finite-sample PPO with clipped objectives cannot provide [5]; CPS supplies this layer at deployment time. Co-design wraps the training environment in CPS, so the actor–critic observes the projected action distribution, whereas shield-at-eval-only policies train without it.

4. Method: Certified Perceptual Shield

CPS is a model-agnostic wrapper around any base policy that proposes $a_k^{\text{raw}} \in \mathcal{B}$. The projection is deterministic: for fixed inputs it always returns the same executed rung. What it certifies is probabilistic, because feasibility is tested against a conformal throughput lower bound that holds with probability at least $1 - \alpha$, where α is the tolerated miss rate. The rest of this section covers the conformal estimator, the banking and feasibility projection (Algorithm 1), a predictive extension, optional PPO co-training, and the evaluation arms.

4.1. Conformal throughput lower bound

The point predictor $\hat{C}_k$ is the harmonic mean of the last κ observed throughputs, RobustMPC-style [3]. After each step, the multiplicative residual $r = C_k^{\text{act}}/\hat{C}_k$ is appended to a rolling window of size W . Given n residuals, the split-conformal α -quantile uses the finite-sample rank $\lfloor \alpha(n+1) \rfloor$ (1-based); the $(n+1)$ term is the standard finite-sample correction. We avoid a linearly interpolated empirical quantile, because it under-covers on short episodes [13, 14]. If that rank is undefined—or while $n < W_{\text{calib}}$ with $W_{\text{calib}}=20$ —we fall back to a fixed scale $\rho_{\text{fb}}=0.80$. That warm-up is not the finite-sample certificate. The lower bound reads $\underline{C}_k = \hat{C}_k \cdot q_\alpha$, clipped to stay at or below $\hat{C}_k$. Under

residual exchangeability, the construction guarantees $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ once the residual window has length at least W_{calib} . Given $\underline{C}_k$, the certified download duration of rung j is

$$d_k(j) = \min \left\{ \frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3 + \varepsilon_0}, 60 \right\}. \quad (5)$$

Proposition 1 (Per-chunk stall certificate). *Fix chunk k , executed index j , buffer $B_k \geq 0.1$ s, and conformal lower bound $\underline{C}_k$ computed with miscoverage α . If $d_k(j) \leq \max\{B_k - m, 0.1\}$ via (5) and the realized throughput satisfies $C_k^{\text{act}} \geq \underline{C}_k$, then $\tau_k^{\text{stall}} = 0$. The buffer floor is required: $\max\{B_k - m, 0.1\} \leq B_k$ only when $B_k \geq 0.1$. Under exchangeability of conformal residuals, $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ at each step where the finite-sample bound is defined ($n \geq W_{\text{calib}}$).*

Proof sketch. When $C_k^{\text{act}} \geq \underline{C}_k$ and $B_k \geq 0.1$ s, the realized download time is at most $d_k(j) \leq B_k$; hence (4) gives $\tau_k^{\text{stall}} = 0$. Coverage follows from standard split-conformal prediction with the $(n+1)$ rank correction [13, 14]. $\square$

Scope of the guarantee. Two parts of the certificate behave differently. The feasibility test is deterministic only when the executed rung satisfies $d_k(j) \leq \max\{B_k - m, 0.1\}$ and $B_k \geq 0.1$ s. If the realized throughput then clears $\underline{C}_k$, that chunk cannot stall. Algorithm 1 can still return rung 0 when $B_k \leq B_{\text{crit}}$, or when the decrement loop reaches 0 while $d_k(0)$ fails the test. Those chunks are outside the certificate. Only the throughput bound is probabilistic. It provides marginal coverage over the exchangeable residual window, $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$, averaged over residual draws rather than per chunk, and only after warm-up. It does not provide coverage conditional on the current buffer or throughput regime. Distribution-free split conformal cannot do so without stronger assumptions [14]. CPS therefore certifies feasibility on every chunk whose bound holds and whose rung passed the test; it does not promise a stall-free session.

Abrupt 5G/mmWave LOS/NLOS and scheduling shifts can break residual exchangeability [6]. Our online defense is a sliding residual window of length W . The recent ratios play two roles: they calibrate q_α , and they are the ratios immediately preceding the chunk we predict. The bound therefore tracks local nonstationarity, even though finite-sample exactness holds only on an exchangeable window. If drift is large, coverage decays. The decay is bounded by a total-variation term between calibration and test residuals [42]. A weighted-conformal variant that down-weights stale residuals is a natural extension, which we leave to future work. Empirically, we treat reported coverage as the operational check of Proposition 1. That rate averages over every chunk with a defined bound, including the W_{calib} warm-up steps that use ρ_{fb} . On the primary 5G pool, the certified arm meets $1 - \alpha$ (0.919 vs. 0.90). Real broadband traces and tighter α can sit marginally below the target (Section 5.3). This is what the theory predicts: short calibration windows and nonstationary traces both erode finite-sample coverage.

4.2. Banking and certified feasibility

Safety set. Indices that never exceed the proposal and fit the buffer margin m form

$$\mathcal{A}_{\text{safe},k}(a) = \{j \leq a : d_k(j) \leq \max\{B_k - m, 0.1\}\}. \quad (6)$$

VMAF-knee banking. The perceptual budget ε is a tolerance in VMAF points. Even when a is feasible, CPS steps down to the smallest rung whose VMAF lies within ε of the proposal, namely

$$j_k^{\text{knee}}(a) = \min\{j \leq a : V_k(a) - V_k(j) \leq \varepsilon\}, \quad (7)$$

The shield then projects $j_k^{\text{knee}}(a)$ onto $\mathcal{A}_{\text{safe},k}(a)$; safety always takes precedence over ε . *Remark.* Banking may run a lower rung than the proposal, and a lower rung can only shorten $d_k(\cdot)$. Once (6) holds, safety projection therefore keeps the certificate intact.

Banking distinguishes CPS from prior VMAF-ranking shields, and the distinction is substantive. On any nondecreasing prefix of a ladder, a ranking shield provably banks nothing.

Proposition 2 (Ranking shields collapse on monotone ladders). *Fix chunk k and suppose the VMAF ladder is nondecreasing in the rung index, $V_k(0) \leq \dots \leq V_k(L-1)$, and that download time $d_k(\cdot)$ is nondecreasing in the rung index. Let $\mathcal{F} = \{j : d_k(j) \leq B_k - m\}$ be the feasible set. Then any shield that returns $\arg \max_{j \in \mathcal{F}} V_k(j)$ returns the highest feasible index $\max \mathcal{F}$, independently of any perceptual-loss budget ε .*

Proof. Because d_k is nondecreasing, $\mathcal{F}$ is a prefix $\{0, \dots, j_{\text{max}}\}$ with $j_{\text{max}} = \max \mathcal{F}$. Since V_k is nondecreasing, $\arg \max_{j \in \mathcal{F}} V_k(j) = j_{\text{max}}$; the budget ε never enters the maximization. $\square$

Ranking by retained VMAF is therefore equivalent to projection onto the highest feasible index, and it frees no bandwidth. CPS instead uses the proposal's quality and steps down to the smallest rung within ε of it (7). This mechanism lets CPS exploit rate-quality saturation: when top rungs carry near-identical VMAF, the knee lies well below j_{max} , and the freed bytes bank as buffer.

Algorithm 1 realizes (7)–(6). If $B_k \leq B_{\text{crit}}$, CPS returns the lowest rung with no feasibility test. Otherwise it computes the knee index from (7) under the effective budget ε_{eff} . This budget equals ε unless predictive mode is on (Section 4.3). CPS then decrements the index until the certified duration (5) respects the margin m , or until the index is 0. If rung 0 still fails the test, the algorithm returns it anyway. The certificate in Proposition 1 does not cover that return, nor the B_{crit} return. The shield never upgrades a proposal. Diagnostics log banked bits, VMAF surrendered, and whether realized throughput met $\underline{C}_k$ (empirical coverage).

Evaluation arms.. RAW: we run a^{raw} unshielded. SAFETY: Algorithm 1 with banking switched off ($j^{\text{knee}}=a$), leaving conformal feasibility alone. CERTIFIED: the full CPS, banking plus feasibility. Paired seeds keep the Wilcoxon and TOST tests valid across arms.

Algorithm 1 Certified Perceptual Shield (compact)

1: **Input:** $a^{\text{raw}}, B_k, \{\sigma_k(j), V_k(j)\}$, conformal estimator; parameters $(\varepsilon, m, B_{\text{crit}})$
2: **Output:** (a^{exec}, ι) with $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$
3: $a \leftarrow \text{clip}(a^{\text{raw}}); \underline{C}_k \leftarrow$ conformal lower bound; define $d_k(\cdot)$ via (5)
4: **if** $B_k \leq B_{\text{crit}}$ **then**
5: **return** $(0, \mathbf{1}[a \neq 0])$
6: **end if**
7: compute ε_{eff} (Section 4.3); $j \leftarrow j_k^{\text{knee}}(a)$ via (7) with ε_{eff}
8: **while** $j > 0$ and $d_k(j) > \max\{B_k - m, 0.1\}$ **do**
9: $j \leftarrow j - 1$
10: **end while**
11: $a^{\text{exec}} \leftarrow \min\{a, j\}$; **return** $(a^{\text{exec}}, \mathbf{1}[a^{\text{exec}} \neq a])$

4.3. Predictive and risk-aware perceptual budget

The predictive budget widens when slack is scarce: it banks lightly when buffer headroom is ample and more aggressively when risk arises. While the buffer stays comfortable, $\varepsilon_{\text{eff}} = \varepsilon$ holds quality steady. Two triggers widen the budget. The first is a drop of B_k below a risk threshold B_{risk} . The second is a short H -step look-ahead under a planning throughput floor (a low quantile of recent observations) that forecasts a dip below B_{risk} ,

$$\varepsilon_{\text{eff}} = \varepsilon + (\varepsilon_{\text{risk}} - \varepsilon) \cdot f(B_k, \text{lookahead}), \quad (8)$$

where $f \in [0, 1]$ ramps between the two budgets and $\varepsilon_{\text{risk}} > \varepsilon$ is the wider risk budget. The shield can therefore pre-bank more deeply, before a stall forces an emergency cut. The predictive-banking experiments (CPS-P) use $\varepsilon=1.0$, $\varepsilon_{\text{risk}}=4$, $B_{\text{risk}}=8$ s, and $H \geq 6$ at forecast quantile 0.2.

4.4. Optional learned hosts and co-design

CPS does not require a learned policy. When the host is an RL policy, we train PPO (optionally Lagrangian-shaped) with Stable-Baselines 3 [5, 43] on content-aware observations: throughput history, buffer, spatial and temporal complexity (SI/TI), and the VMAF ladder. *Co-design*: wrapping the training environment in CPS changes the action distribution observed by the actor-critic. The policy can then learn to propose bolder rungs that banking trims for safety. To gauge the effect, we compare two policies under identical 5G train/test provenance: one trained without the shield (shield-at-eval-only), one co-trained with CPS. Both face CPS-P at test time.

4.5. Implementation notes

At the primary 5G operating point the defaults are $\alpha=0.10$ (coverage target 0.90), residual window $W=200$, warm-up $W_{\text{calib}}=20$, fallback $\rho_{\text{fb}}=0.80$, predictor $\kappa=5$, margin $m=0.5$ s, $B_{\text{crit}}=0.3$ s, and buffer cap 12 s. Each episode has 48 chunks and the residual window resets at the episode boundary, so $W=200$ never fills. The first W_{calib} chunks of every episode use ρ_{fb} . Table 4 collects the shield and predictive-banking defaults from the primary experiments. Empirical coverage is the fraction of steps with $C^{\text{act}} \geq \underline{C}_k$, counted over all chunks including warm-up. Per decision, the projection runs in $O(L+H)$ over the L ladder rungs and an H -step look-ahead. Conformal-quantile upkeep on the length- W residual window is $O(1)$ amortized. Across 20,000 rollout decisions on a commodity CPU ($L=6$, full predictive shield), per-decision wall time is 131 μs at the median

Table 4: CPS default hyperparameters (primary 5G suite).

Parameter	Value
Perceptual budget ε	1.0 VMAF point
Conformal miscoverage α	0.10 (target coverage 0.90)
Residual window W / warm-up W_{calib}	200 / 20
Fallback scale ρ_{fb} / predictor κ	0.80 / 5
Buffer margin m / critical B_{crit}	0.5 s / 0.3 s
Risk budget $\varepsilon_{\text{risk}}$ / B_{risk}	4.0 / 8 s
Look-ahead horizon H / forecast quantile	≥ 6 / 0.2

and 310 μs at the 99th percentile. This is roughly 10^{-5} of the 4 s chunk budget. CPS therefore runs inline, and the control loop is unaffected.

5. Experimental evaluation

We organized the evaluation around four research questions:

- RQ1** *Banking vs. safety.* Does VMAF-knee banking reduce bitrate and rebuffering relative to conformal safety-only projection, with VMAF inside $\pm\varepsilon$?
- RQ2** *Coverage.* Does empirical conformal coverage meet the finite-sample target $1-\alpha$?
- RQ3** *Budget extensions (supporting).* Relative to the default budget ε , do CPS-P or a larger static ε change bitrate and rebuffering savings?
- RQ4** *Robustness.* Do the gains of RQ1–RQ3 persist on the non-monotone per-chunk ladder? Do they remain stable as ε and α vary?

A supporting single-seed co-design study appears in Section 5.4; no certified claim rests on it.

5.1. Experimental setup and datasets

The testbed is a multi-video, chunk-timescale emulator ($K_e=48$ chunks/episode, $T_c=4$ s, buffer cap 12 s unless noted). It drives twelve publicly available reference titles with session-mean VMAF ladders [9]: five Blender open animations (Big Buck Bunny, Tears of Steel, Sintel, Elephants Dream, Cosmos Laundromat) and seven natural-motion Xiph DERF clips (Park Joy, Old Town Cross, Rush Hour, Into Tree, Sunflower, Kristen & Sara, Ducks Take Off). We selected them for ladder structure, not catalogue breadth: saturation-rich animation (Big Buck Bunny, Tears of Steel), steep natural motion (Park Joy, Ducks Take Off), wide-span DERF clips (Into Tree, Old Town Cross), and mixed held-out titles (Sintel, Cosmos Laundromat, Kristen & Sara) for generalization checks (Table 3). Nine titles appeared in PPO co-design training; three were held out entirely (Sintel, Cosmos Laundromat, Kristen & Sara). The negative broadband result (Section 5.3) showed a regime in which banking yields little benefit. CPS is a deterministic per-chunk projection fixed by the observed VMAF ladder and buffer state, so ladder shape—not semantic content—drove its effect. The twelve-title suite therefore spanned both saturated and steep/inverted session-mean ladders (67% non-monotone).

Primary 5G suite. The primary pool used synthetic 5G traces with intermittent dips and outages, held out from the co-design training seeds (204 paired episodes). Public per-chunk 5G/mmWave logs suitable for chunk-level ABR replay remain scarce, so we generate this pool and tune its dip-and-outage statistics to reported 5G/mmWave measurements [6, 21]. The broadband suite below provides a real-trace counterpart. Train and test trace pools stayed source-disjoint. Co-design policies trained on the training pool alone and faced held-out test seeds.

Broadband suite. The broadband suite used real-measured traces: the FCC Measuring-Broadband-America fixed-line logs [16] and the Norway/Nor-net mobile-broadband mobility logs [17] (27 held-out test traces, source-disjoint from training). Both are widely used public datasets in ABR research. They provided a real-network counterpart to the primary 5G suite and showed where banking offered limited headroom.

Hosts. Five host families covered the ABR design space: bitrate-greedy (always at the top rung), buffer-based ABR (BBA-like), BOLA (the Lyapunov buffer-based rule of the dash.js reference player [2, 44]), robust model-predictive control (RobustMPC [3]), and PPO policies (content-aware “proposed” and content-blind Pensieve-style). RobustMPC was a five-segment planner over a throughput prediction discounted by recent worst-case error. CPS carried no host bias: one wrapper, applied at inference across all five families, tuned to none of them.

Arms and pairing. Every host rolled out RAW, SAFETY, and CERTIFIED on identical Python seeds—the same video–trace pairs. Pairing fixes the random seed and the (video, trace) assignment across arms, so metrics differ only because of shielding, not because of resampling. Each paired contrast is the per-episode metric difference on the same (video, trace) seed across arms (e.g., CERTIFIED minus SAFETY on episode i). We reported mean delivered bitrate, mean rebuffering per episode, stall-free rate, mean VMAF, intervention rate, and empirical conformal coverage. Significance rests on paired Wilcoxon signed-rank tests (zero-method `zsplit`) and on two one-sided tests (TOST) for equivalence of VMAF within $\pm\epsilon=1.0$ [15]. Defaults were $\alpha=0.10$ and coverage target 0.90. We set $\epsilon=1.0$ as both the knee budget and the TOST margin. One VMAF point is a small fraction of typical adjacent-rung and top-rung gaps on our ladders (Table 3) and is conservative relative to perceptually aware per-title encoding practice [40]. It is an engineering design choice aligned with (7), not a preregistered clinical equivalence margin [15]. Because the knee rule and this TOST both use the session-mean ladder, a mean change inside $\pm\epsilon$ is partly by construction. The per-chunk ablation checks the true chunk VMAF instead (Section 5.5). We report bitrate, rebuffering, and VMAF separately. The scalar in (2) is the logging definition. Only the co-design sweep uses the linear score $QoE(w)$ defined in Section 5.4. The p -values we quoted for the primary arms were raw and two-sided Wilcoxon signed-rank tests (zero-method `zsplit`). Identical p -values across hosts can arise when two contrasts share the same signed-rank statistic. The co-design values 5.0×10^{-5} and 5.0×10^{-5} equal $1/20,000$, the floor of a 20,000-draw sign-flip test. They are not signed-rank tail probabilities. The bootstrap intervals in Section 5.4 are the magnitude statement for that contrast. Since several paired contrasts ran at once, we checked that

every reported bitrate and rebuffering effect on the primary arms survived Holm–Bonferroni correction at 0.05 over the reported family. One caveat concerns content diversity. Each episode is one (video, trace) pair: the video is drawn uniformly from the twelve titles and the trace from the held-out pool. Observations therefore clustered by title, and effective independent-content diversity was twelve, not 204. We treat the reported p -values as within-pair evidence on that paired design, not as evidence from 204 independent titles.

5.2. Model-agnostic banking on 5G (RQ1–RQ2)

Table 5 reports absolute metrics for all three arms—RAW, SAFETY, and CERTIFIED—on the primary 5G test traces for five host families: greedy, BBA, BOLA, RobustMPC, and a Pensieve-style negative control. Fig. 3 distills only the certified-vs.-safety banking contrast in bar form (not RAW); the table lists the underlying episode means and significance tests for every arm. Banking behaved consistently across all four deterministic hosts on the twelve-title suite. Certified banking vs. safety saved 4.5%, 4.1%, 4.1%, and 3.8% bitrate on greedy, BBA, BOLA, and RobustMPC, respectively, and reduced rebuffering by 9.3%, 8.9%, 8.9%, and 9.2% (Table 5; greedy $p=8.5\times 10^{-4}$). On steep, non-saturated ladders (Table 3) the absolute savings stay modest. This is what the mechanism predicts: banking pays off only where the ladder saturates. Every host nevertheless remains perceptually near-lossless. Partitioning the greedy certified–safety contrast by session-mean top-rung VMAF gain (Table 3) made that dependence explicit. On titles with top gain at most 2 VMAF points (Big Buck Bunny, Sintel, Sunflower, Kristen & Sara; $n=60$ episodes) banking saved 4.6% bitrate and 9.9% rebuffering. On steep titles with top gain at least 9 (Park Joy, Ducks Take Off, Into Tree; $n=51$) the same contrast shrank to 2.1% bitrate and 3.7% rebuffering. The pooled greedy headline (4.5% bitrate saved and 9.3% rebuffering reduced) therefore mixed both regimes rather than reflecting a saturation-only subset. Mean VMAF differences were perceptually negligible, at -0.01 , 0.17 , 0.18 , and 0.02 points across greedy, BBA, BOLA, and RobustMPC, with every value inside $\pm\epsilon$ by TOST. The two shield layers split the work. On greedy, mean rebuffering per ~ 192 s episode (48 chunks) falls from 85.5 s (RAW) to 41.9 s (SAFETY) and 38.0 s (CERTIFIED). Against RAW, certified greedy reduced rebuffering by 55.5% (BBA: 51.0%). Conformal safety removed most of the stall time relative to RAW; banking added the certified–safety margin on top. Fig. 5 and Fig. 6 show per-episode operating points and rebuffering CDFs for greedy and BBA. Coverage followed a distinct pattern on the primary 5G pool. Empirical coverage there was 0.919 (greedy) and 0.919 (BBA) against a 0.90 target, at or above the nominal level on this pool, consistent with the $(n+1)$ conformal calibration at these episode lengths (Section 4.1). Because coverage depends on the throughput trace and the conformal calibrator rather than the base policy, it was identical (0.919) across every host and shield variant on the 5G test set (Fig. 4). Proposition 1 is a per-chunk probabilistic certificate, not a per-episode stall-free one. On the greedy certified arm we attribute every second of residual rebuffering on the primary 5G pool (204 paired episodes) to an exclusive cause. When the residual window is shorter than the

conformal warm-up ($W_{\text{calib}}=20$), 91.3% of that stall time occurs; the bound then uses the fixed fallback scale and is not the finite-sample certificate. After warm-up, 8.7% falls on chunks where realized throughput misses $\underline{C}_k$. 0.0% falls on chunks where the bound holds, matching Proposition 1. The $B_k \leq B_{\text{crit}}$ emergency path contributes 0.0% of stall seconds on this suite. Pensieve, the negative control, degraded to near-floor bitrates with near-zero stalls, so CPS rarely intervened. Banking could not rescue a policy that never requested saturated rungs.

The intervention rate under the greedy host is high by construction (Table 5). It does not imply that the shield dominates an informed base policy. Greedy proposed the top rung every time, so nearly every banking downshift or feasibility repair appeared as an intervention. The rate gauged how far a host sat from the certified operating point, not how much of the outcome to attribute to the shield. This is why we ran the model-agnostic contrast across hosts spanning the spectrum.

The certified intervention rate falls as the host becomes more conservative: 70.4% (greedy), 66.1% (BBA), 61.1% (BOLA), 31.1% (RobustMPC), and 7.4% (Pensieve-style). The sharper the base policy, the lighter the shield’s load. Coverage nevertheless holds at 0.919 throughout. The certified metrics therefore depend increasingly on the base policy’s own decisions, and the shield acts only where banking headroom remains. For RQ1 and RQ2 the message is simple: on every deterministic host, banking saves bitrate and cuts rebuffering with mean VMAF inside $\pm\epsilon$, and empirical coverage meets the 0.90 target on the primary 5G pool when warm-up chunks that use ρ_{fb} are included in the average.

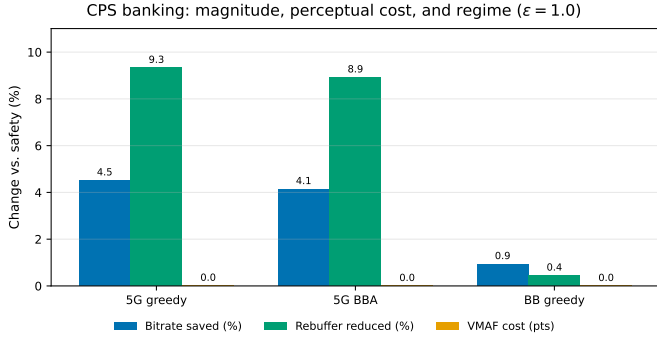


Figure 3: CPS banking overview (certified vs. safety, 204 paired episodes, $\epsilon=1.0$). Bars show percent bitrate saved, percent rebuffering reduced, and mean VMAF difference in points (CERTIFIED minus SAFETY) for two primary-5G hosts and one broadband host.

5.3. Predictive banking and broadband boundary (RQ3)

With risk-aware $\epsilon(B)$ and dip forecasting enabled (CPS-P; budget widening up to $\epsilon_{\text{risk}}=4$ VMAF points), certified banking vs. safety reaches a 14.6% bitrate reduction (greedy) and 12.5% (BBA), and cuts rebuffering by 12.6% and 13.0%, respectively. These gains are relative to the default budget $\epsilon=1.0$. A static budget $\epsilon=2$ on the same greedy host already saves 14.4% bitrate and 30.8% rebuffering, at a mean VMAF change of -0.17 (Table 10). CPS-P therefore matches that bitrate saving but does not improve on the static rebuffering cut. Its role is to keep ϵ at the base value while the buffer is comfortable, and to widen it

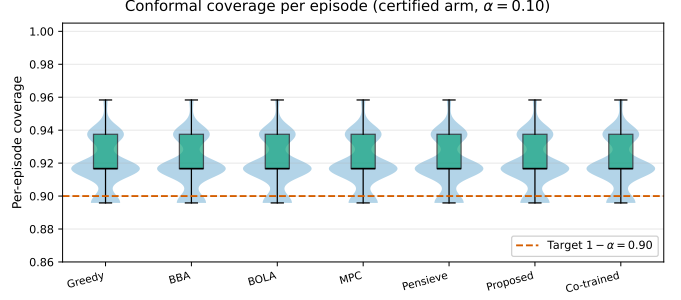


Figure 4: Per-episode conformal coverage of the certified arm on the primary 5G test pool ($\alpha=0.10$, target 0.90, dashed). The distribution is host-invariant on this pool (RQ2).

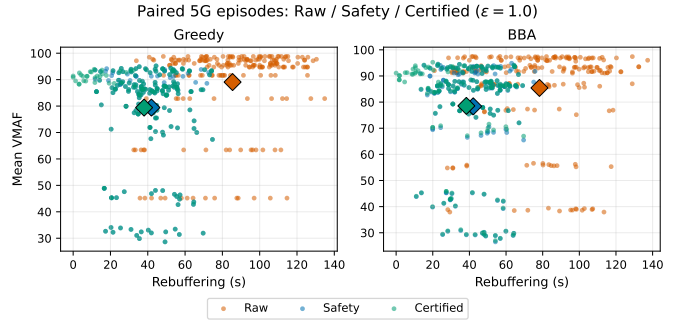


Figure 5: Per-episode operating points on the primary 5G suite (paired seeds). Diamonds mark arm means (RAW, SAFETY, CERTIFIED).

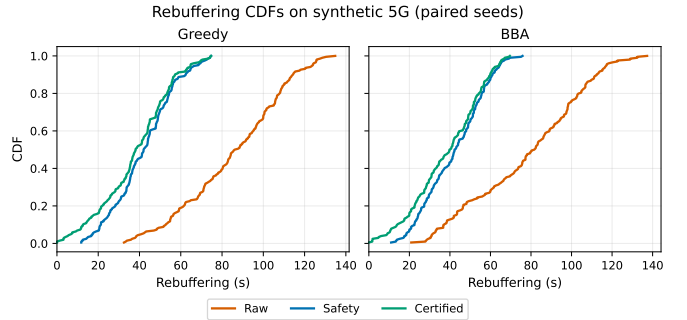


Figure 6: Empirical CDFs of episode rebuffering for greedy and BBA hosts. The certified arm’s rebuffering distribution lies at or left of the safety arm and well left of raw.

Table 5: Paired arms on primary 5G test traces (204 episodes, $\epsilon=1.0$, $\alpha=0.10$). Reb. is mean rebuffering per episode; Cover. is empirical conformal coverage; Interv. is shield intervention rate.

Host	Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Greedy	RAW	85.5 s	89.1	6000	—	0.0
	SAFETY	41.9 s	79.4	3131	0.919	65.8%
	CERTIFIED	38.0 s	79.4	2990	0.919	70.4%
BBA	RAW	78.2 s	85.4	4662	—	0.0
	SAFETY	42.1 s	78.4	2809	0.919	58.1%
	CERTIFIED	38.3 s	78.5	2692	0.919	66.1%
BOLA	RAW	77.6 s	83.2	4627	—	0.0
	SAFETY	42.4 s	77.5	2782	0.919	54.2%
	CERTIFIED	38.7 s	77.7	2667	0.919	61.1%
RobustMPC	RAW	71.3 s	77.4	3175	—	0.0
	SAFETY	40.0 s	74.9	2377	0.919	28.3%
	CERTIFIED	36.3 s	74.9	2287	0.919	31.1%
Pensieve	RAW	0.1 s	72.2	722	—	0.0
	SAFETY	0.1 s	70.8	689	0.919	7.3%
	CERTIFIED	0.1 s	70.8	689	0.919	7.4%

Certified–safety Wilcoxon: greedy bitrate $p=8.5\times 10^{-4}$, rebuffering $p=8.5\times 10^{-4}$; BBA bitrate $p=4.2\times 10^{-16}$, rebuffering $p=2.5\times 10^{-4}$. CPS-P predictive banking defaults in Table 4.

Table 6: Paired greedy arms on real broadband traces (204 episodes over 27 held-out FCC/Norway traces, $\epsilon=1.0$, $\alpha=0.10$). Certified vs. safety saves 0.9% bitrate; safety vs. raw cuts rebuffering by 98.7%.

Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
RAW	1216.1 s	89.08	6000	—	0.0%
SAFETY	16.2 s	69.58	1180	0.890	97.5%
CERTIFIED	16.2 s	69.58	1169	0.890	97.9%

only under risk. Mean VMAF changes under CPS-P are -0.23 and -0.02 points.

The broadband suite marked the edge of these gains. On source-disjoint real-measured broadband traces (FCC MBA and Norway/Nornet [16, 17]; 27 held-out traces, 204 paired video-trace episodes), greedy certified–safety bitrate savings shrank to 0.9% (Table 6). Top rungs were seldom jointly feasible and saturated, so banking had little remaining headroom. Even so, safety vs. raw still reduces rebuffering by 98.7%. Raw greedy already stalls for most of each broadband session (Table 6). The cut is what a feasibility projection does against that baseline. It is not evidence that the conformal bound, as opposed to any conservative rung cut, transfers across regimes. On these 27 held-out traces at $\alpha=0.10$, empirical conformal coverage was 0.890 (below the 0.90 target). The primary 5G pool, over 204 episodes, reached 0.919 (Table 5, Fig. 4). The “BB greedy” group in Fig. 3 made the regime split explicit: banking savings were largest on the primary 5G suite. This split was by design: CPS banking paid off precisely when the ladder knee sat within reach, and broadband placed it out of reach.

Coverage under injected drift. To stress-test exchangeability without new traces, we replayed the primary 5G pool with a mid-episode throughput shock (chunks 28–37: multiply trace throughput by 0.3, mimicking a sudden loss of line-of-sight (LOS) or a non-line-of-sight (NLOS) dip). We report two related

Table 7: Paired arms for the primary PPO host on the 5G test pool (broadband-trained policy, 204 episodes, $\epsilon=1.0$, $\alpha=0.10$). Certified vs. safety: -35.3% bitrate, -79.3% rebuffering.

Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
RAW	3.90 s	76.42	1589	—	0.0%
SAFETY	2.96 s	72.99	1278	0.919	18.4%
CERTIFIED	0.61 s	72.66	827	0.919	67.2%

views. First, phase-mean coverage over valid chunks after a 20-chunk conformal warm-up: 0.928 pre-shock ($n=1,632$), 0.929 during the shock ($n=2,040$), and 0.944 post-shock ($n=2,040$), all against a $1-\alpha=0.90$ target (greedy certified arm, 204 episodes). Those phase averages therefore stayed at or above the target through the shock. Second, Fig. 7 plots the episode-averaged rolling coverage with window $W=20$ chunks. Because the window still contains cold-start residuals, the rolling curve starts below the target and only crosses 0.90 near the end of the shock band; the shaded warm-up marks that ramp. The sliding residual window then keeps phase-mean coverage on target after calibration, consistent with Section 4.1.

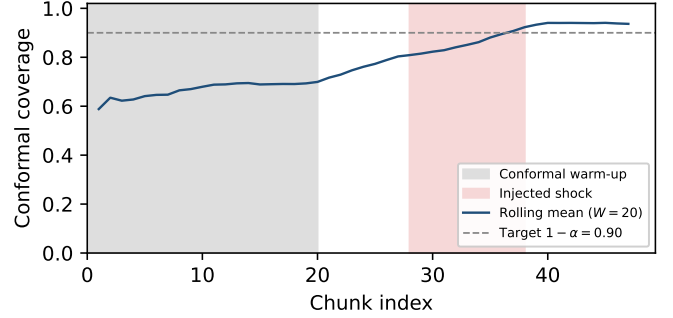


Figure 7: Episode-averaged rolling conformal coverage ($W=20$ chunks) under an injected throughput shock (red band; 0.3 $\times$ scale, chunks 28–37) on the primary 5G pool. Grey band: conformal warm-up (chunks 0–19). Dashed line: target $1-\alpha=0.90$.

5.4. Learned hosts and co-design

We report two PPO settings separately. The primary host was a content-aware policy trained on source-disjoint broadband traces and evaluated, without further training, on 5G—the model-agnostic deployment case (Table 7). The co-design pair comprised two policies trained on matched 5G traces—one shield-at-eval-only, one co-trained with CPS in the loop—compared on their certified arms (Table 8 and Fig. 8). On the primary broadband-trained policy, certified banking vs. safety saved 35.3% bitrate and reduced rebuffering by 79.3% ($p=6.2\times 10^{-18}$), at VMAF $\Delta = -0.33$ (TOST within $\pm\epsilon$). The bitrate cut exceeds that of greedy or BBA. Table 7 shows why. Raw mean bitrate is 1589 kb/s, not the top rung. Safety already lowers it to 1278 kb/s. Certified banking then lowers it to 827 kb/s. The 35.3% figure is that second cut, not evidence that the raw policy proposed saturated rungs. Under CPS-P, banking saved 21.9% bitrate and reduced rebuffering by 29.6%, mirroring the predictive-banking gains on deterministic hosts in Section 5.3.

Co-design. Both policies trained on matched 5G traces (source-disjoint test). Under CPS-P at eval, the co-trained policy’s certified arm attained VMAF 75.73 and bitrate 1463 kb/s,

Table 8: Co-design on 5G test (certified arm, CPS-P at eval).

Policy	VMAF	Reb. (s)	Bitrate	Cover.
Eval-only shield	72.80	0.31	801	0.919
Co-trained (CPS)	75.73	0.86	1463	0.919
Δ (co – eval)	+2.93	+0.55	—	—

The Δ row equals the certified-arm difference in the rows above. Paired Wilcoxon: VMAF $p=5.0\times 10^{-5}$, rebuffering $p=5.0\times 10^{-5}$; bootstrap Δ VMAF CI [2.61, 3.25] over 204 episodes.

against 72.80/801 for shield-at-eval-only operation. The Δ row in Table 8 was the difference of those certified-arm means (+2.93 VMAF, +0.55 s rebuffering), and it matched the raw rows above (75.73 – 72.80, 0.86 – 0.31). The fidelity gain could be weighed against its stall cost through $\text{QoE}(w) = \text{VMAF} - w \cdot \text{rebuffer}(s)$. The mean-QoE crossover was $w^*=5.31$ VMAF points per second (ratio of the Δ row entries): co-design favored fidelity-oriented weights $w < w^*$; shield-at-eval-only prevailed where stall aversion was stronger. Intuitively, w^* is the stall penalty at which the co-trained fidelity gain and extra rebuffering exactly balance in mean QoE; weights above w^* favor the shield-at-eval-only policy. The reported figures 5.0×10^{-5} and 5.0×10^{-5} are both 5.0×10^{-5} . That value is 1/20,000, the floor of the sign-flip test, not a Wilcoxon tail probability. The bootstrap interval [2.61, 3.25] is the magnitude statement. The bootstrap CI was [2.61, 3.25], with stall cost [0.29, 0.85] s and crossover [3.40, 10.27] (median 5.37; positive VMAF in 100.0% of resamples). Co-design remained a supporting result under a single training seed (Section 6).

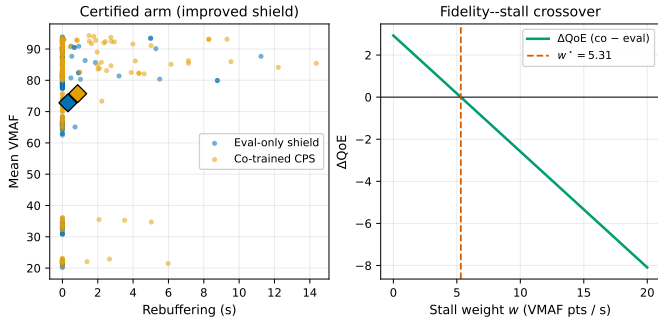


Figure 8: Co-design on 5G test (certified arm, CPS-P). Left: per-episode VMAF–rebuffer points (diamond markers = episode means). Right: mean $\Delta\text{QoE}(w)$ vs. stall weight; the dashed line marks the crossover $w^*=5.31$.

5.5. Per-chunk (non-monotone) ladder ablation (RQ4)

The headline runs applied the knee rule to each title’s pooled session-mean ladder (Table 3): four of twelve ladders are fully nondecreasing, while the other eight are non-monotone only at mid-ladder (750→1200 kbps); the top three rungs remain non-decreasing on every title. We therefore tested whether banking remained effective on the true per-chunk ladders, which were often non-monotone (26.5% of the 113 chunks held at least one inverted adjacent pair; Fig. 2). The per-chunk ablation uses the four titles with frame-level VMAF logs (Big Buck Bunny, Tears of Steel, Sintel, CrowdRun), the same family as Fig. 2. CrowdRun is not in the twelve-title session-mean roster of Table 3. The ablation reads 113 chunks from the aggregated per-chunk export; the figure aggregates 173 chunks from raw per-frame

Table 9: Banking-stage ablation on real per-chunk VMAF ladders (greedy proposal, 113 chunks). A per-chunk shield banks slightly more bytes than the pooled shield while keeping realized per-chunk cost $\leq \epsilon$ by construction; the pooled shield incurs a small per-chunk budget-violation rate. Feasibility is VMAF-independent and unchanged.

ϵ	Mean knee rung		Bitrate saved (%)		Per-chunk cost (pts)	Budget viol. pooled (%)
	pooled	per-chunk	pooled	per-chunk	per-chunk	
0.5	4.58	4.34	22.3	28.1	0.02	8.8
1.0	4.15	3.94	44.6	45.8	0.32	11.5
2.0	3.60	3.39	58.2	59.7	0.98	18.6
4.0	2.93	2.88	68.7	68.7	1.89	7.1

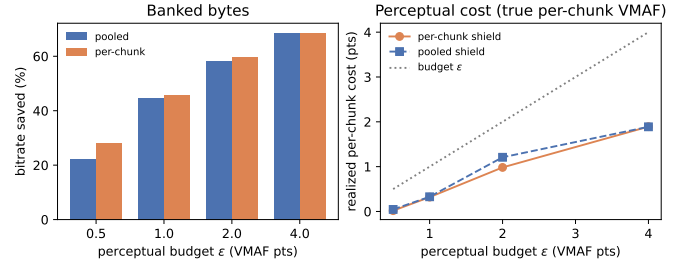


Figure 9: Per-chunk vs. pooled ladder banking across the perceptual budget ϵ . Left: banked bitrate for both shields. Right: realized per-chunk perceptual cost for both shields against ϵ .

JSON, so the two counts differ. To test this, we replay the deterministic banking stage on the per-chunk logs. We compare a shield that sees only the pooled ladder with one that sees the per-chunk ladder (Table 9, Fig. 9). Banking alone depended on VMAF; the feasibility projection ignored it and stayed identical in both cases, which isolated the effect of the pooling simplification. Three findings held at the default $\epsilon=1.0$, spanning correctness, savings, and pooling error. First, the knee rule remained well defined on inverted ladders and banked more, not less: mean saved bitrate was 45.8% per-chunk against 44.6% pooled, because a lower rung that inverted above the proposal satisfied the budget immediately—inversions widened banking opportunity rather than breaking the rule. Second, a per-chunk shield held realized per-chunk perceptual cost inside budget by construction (mean 0.32 VMAF points, never above ϵ), so the ablation validated chunk-wise enforcement when true ladders were available. Third, pooling was not free. Judged against the true per-chunk VMAF, the pooled-ladder shield overran the budget on 11.5% of chunks (mean cost 0.33 points), which explains why session-mean evaluation could understate perceptual spend even when banking bytes looked similar. Taken together, the pooled-ladder headline numbers were, if anything, conservative with respect to banking savings. Per-chunk operation was sound: it eliminated residual budget violations at the cost of a per-chunk VMAF feed, and it aligned the content-diversity motivation with the pooled evaluation.

5.6. Sensitivity to ϵ and α (RQ4)

The headline runs fixed the pair at $\epsilon=1.0$ and $\alpha=0.10$. Table 10, Table 11, and Fig. 10 vary one parameter at a time on the greedy host and the primary 5G pool (204 paired episodes; twelve-title roster). The default cell ($\epsilon=1.0$, $\alpha=0.10$) is the

Table 10: Sensitivity to the perceptual budget ε (greedy host, 5G test, $\alpha=0.10$, certified vs. safety). The $\varepsilon=1.0$ row is the headline evaluation in Table 5; other rows vary ε only. Banking scales smoothly with ε ; the mean VMAF change stays negligible and coverage is invariant to ε .

ε (pts)	Bitrate saved (%)	Rebuffer Δ (%)	VMAF Δ (pts)	Coverage
0.5	2.0	-3.5	+0.02	0.919
1.0	4.5	-9.3	-0.01	0.919
2.0	14.4	-30.8	-0.17	0.919
4.0	24.6	-46.1	-0.48	0.919

Table 11: Sensitivity to the conformal miscoverage α (greedy host, 5G test, $\varepsilon=1.0$). The $\alpha=0.10$ row is the headline evaluation in Table 5; other rows vary α only. Empirical coverage tracks the finite-sample target $1-\alpha$, and the intervention rate rises as α tightens.

α	Target $1-\alpha$	Coverage	Interv. (%)	Bitrate saved (%)
0.05	0.95	0.945	73.0	4.5
0.10	0.90	0.919	70.4	4.5
0.20	0.80	0.854	35.2	6.2

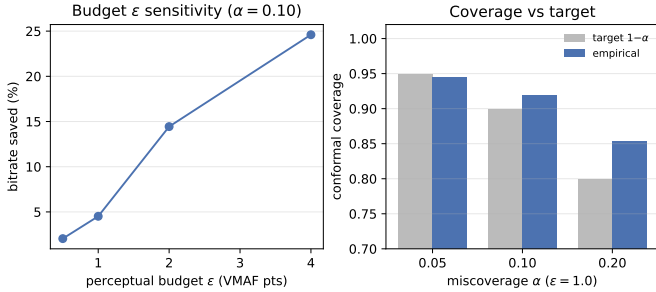


Figure 10: Hyperparameter sensitivity. Left: banked bitrate rises and the mean VMAF change stays near zero as the budget ε grows ($\alpha=0.10$). Right: empirical conformal coverage tracks the target $1-\alpha$ across α ($\varepsilon=1.0$).

same evaluation as Table 5; the other cells are a one-parameter sensitivity sweep under the same seeding protocol. We first considered the perceptual budget. Sweeping $\varepsilon \in \{0.5, \dots, 4.0\}$ (at $\alpha=0.10$) traded banking for fidelity. Bitrate saved vs. SAFETY climbed from 2.0% to 24.6%, while the mean VMAF change remained negligible (at most 0.48 points) and never left the matching $\pm\varepsilon$ envelope—so wider budgets traded fidelity slack for banking headroom in a predictable way. Empirical conformal coverage remained invariant with ε . Coverage depended on the throughput trace and calibrator alone, not the banking budget, which showed that banking and the feasibility certificate operated independently by design. We next considered miscoverage. Sweeping $\alpha \in \{0.05, 0.10, 0.20\}$ (at $\varepsilon=1.0$) on the same primary 5G pool moved empirical coverage in step with the finite-sample target $1-\alpha$: from 0.945 at target 0.95 through the default 0.919 at target 0.90 to 0.854 at target 0.80. Tightening α raises both coverage and the intervention rate; loosening it lowers both, as the $(n+1)$ construction predicts. At $\alpha=0.05$ the mean was 0.945, marginally below the 0.95 target at $n=204$ (Section 5.3). The default operating point therefore lay on a smooth, well-behaved frontier rather than a narrowly tuned optimum, with headline savings stable once ε and α were fixed.

6. Discussion and limitations

Scope. CPS is an auditable, model-agnostic mechanism with two stages. First, it banks bandwidth on VMAF-saturated ladder knees inside a stated perceptual budget. Second, it enforces download feasibility under a conformal throughput lower bound with target coverage $1-\alpha$, where α is the tolerated miss rate. We state the limits of the claim plainly. CPS does not claim dominance over every ABR controller. Its feasibility guarantee is probabilistic rather than deterministic: coverage holds at the $1-\alpha$ level when calibration residuals are exchangeable, and we claim no pathwise worst-case bound. The benefit also depends on the network regime and ladder shape. On the primary 5G suite, conformal safety already cuts mean greedy rebuffering from 85.5 s to 41.9 s per ~ 192 s episode; certified banking trims it further to 38.0 s. Banking gains accrue where session-mean ladders saturate at the top and the host requests those rungs, not because high rungs are jointly feasible on every chunk. Under broadband stress, the safety projection dominates and banking headroom decreases.

Local ladder inversions. Fig. 2 shows that, chunk by chunk, a sizable share of adjacent rungs in titles such as Sintel and Big Buck Bunny run near-flat or inverted. A higher bitrate then scores no better, and sometimes worse, in VMAF. CPS handles all of this without a special case. The headline shield uses each title’s session-mean ladder (Table 3). Proposition 2 therefore applies in full only on the four fully nondecreasing titles. On the eight titles with mid-ladder inversions, it still applies to the upper three rungs, where greedy banking concentrates. Even on a fully per-chunk, non-monotone ladder, both stages remain well defined. Each stage, moreover, handles inversions on its own terms. The feasibility projection (6) uses download time, never VMAF, so inversions leave safety untouched. The knee rule (7) picks the smallest rung within ε of the proposal’s VMAF. When an inversion lets a lower rung score at least as high, that rung satisfies the budget on its own, and banking captures the extra headroom rather than failing on it. Local inversions, then, can only widen the banking opportunity, and the algorithm handles them by construction. The per-chunk ablation in Section 5.5 confirms this: banked bytes match, and realized per-chunk perceptual cost stays inside budget. Per-chunk ladders improve accuracy; they are not required for correctness.

Threats to validity.

1. The broadband suite replays real-measured FCC and Norway/Norinet traces [16, 17], whereas the 5G suite is synthetic. In both cases the chunk-level emulator omits transport-stack dynamics and CDN caching [6]. We leave live-player validation to future work.
2. Twelve publicly available titles and finite trace pools still cap catalogue diversity [17]; three titles are held out from PPO training but recur in CPS evaluation.
3. QoE here is a VMAF-based surrogate. Subjective rankings of stall against fidelity may diverge from it [8].
4. Conformal coverage presumes exchangeable residuals (Section 4.1). Empirical coverage can sit below the nominal $1-\alpha$ target on the 27 broadband traces (0.890 at $\alpha=0.10$),

or leave only a tight miscoverage margin on the primary 5G suite (0.945 vs. 0.95 at $\alpha=0.05$, $n=204$; Table 11). Headline RQ2 reports the default 5G operating point (0.919 vs. 0.90).

5. Every arm—greedy, BBA, the PPO hosts—runs locally under the same finite-sample conformal correction (Section 4.1). Co-design magnitudes hinge on the checkpoint, so we report the best-validation model of each policy. The Δ rows of Table 8 come from certified-arm means. The paired Wilcoxon test and bootstrap intervals use per-episode exports over the same 204 pairs, with both co-design policies re-evaluated together. The co-design contrast uses a single training seed per policy. The 95% intervals therefore reflect evaluation variance only, not training-seed variance. Establishing training-seed variance for the crossover weight w^* would require retraining many seeds, which we leave to future work. Co-design therefore stands as a supporting result, not a headline claim.
6. Banking savings scale with ladder saturation (Table 3; see also the top-gain split in Section 5.2). The twelve-title mix deliberately includes steep titles, so pooled effect sizes are conservative relative to a saturation-rich subset alone.
7. The Pensieve-style no-op shows that host aggressiveness sets the banking opportunity. Results therefore do not transfer to arbitrary policies without the paired-arm protocol.
8. The margin for the two one-sided tests (TOST) equals the configured banking budget ε (Section 5.1); we justify $\varepsilon=1.0$ relative to ladder spacing and perceptual encoding practice [40], but it remains a design parameter rather than a clinical equivalence margin [15].

Future work. Several directions remain open: packet-level or field 5G evaluation; applying CPS to RobustMPC and Comyco hosts; conditional value-at-risk (CVaR) and other tail-aware objectives [45]; telemetry-driven adaptation of ε and α ; and preregistered non-inferiority studies with a fixed perceptual margin. Real-player deployment is the highest-priority direction. Every result here comes from a chunk-level emulator that omits transport-stack dynamics. Inserting CPS into a production player such as `dash.js` [44] therefore raises concrete challenges that the simulation never sees. Three challenges are central. First, throughput must be estimated from noisy, bursty HTTP/TCP (or QUIC) samples. Their exchangeability—the assumption behind the conformal bound—can break under slow-start, congestion-control transients, and CDN re-routing. The calibration residuals then require periodic re-estimation, or a sliding, weighted-conformal variant that provably bounds coverage loss under drift [42]. Second, the feasibility check must reconcile the player’s own buffer-occupancy signal, which is segment-level and delayed, with the certificate’s buffer-margin accounting. Third, the per-decision budget tightens from our $\sim 100 \mu\text{s}$ microbenchmark into a soft real-time constraint running beside the player’s ABR loop. Showing that the finite-sample coverage guarantee withstands these real-world perturbations—and measuring any coverage slack they open up—is the natural next step.

7. Conclusion

We present the Certified Perceptual Shield (CPS), a conformal shielding layer for client-side ABR. CPS combines three parts: bandwidth banking at the VMAF knee, an online conformal throughput lower bound with finite-sample coverage under residual exchangeability, and a certified buffer-feasibility test. We evaluate CPS on 204 paired 5G episodes from twelve titles with diverse ladders. Relative to the safety-only conformal projection, banking lowers delivered bitrate by 3.8–4.5% and rebuffering by 8.9–9.3%. This holds for every deterministic host, from greedy to RobustMPC. Mean VMAF stays inside the tolerance ± 1.0 VMAF points. Empirical coverage of the throughput lower bound is 0.919, against a target of 0.90, including warm-up chunks that use the fallback scale. Optional CPS-P and static budget sweeps (Section 5.3) show that widening ε via look-ahead or a fixed larger value can deepen bitrate savings; static $\varepsilon=2$ still cuts greedy rebuffering more than CPS-P at the default budget. Two regimes therefore appear. Banking supplies the savings while the ladder knee lies inside the observable operating range. Under broadband stress the knee moves beyond that range, and a feasibility projection—not a transferred conformal bound—accounts for the stall cut against raw greedy. A supporting single-seed experiment co-trains PPO with CPS in the loop (Section 5.4). It gains +2.93 VMAF at +0.55 s of rebuffering, with a QoE crossover weight $w^*=5.31$. CPS attaches to an existing ABR controller as a lightweight wrapper and needs no retraining, and it can also wrap a learned policy. Spare perceptual quality therefore becomes a budget that the client can measure and spend under a finite-sample guarantee.

Appendix A. Lagrangian CMDP training for RL hosts

An unconstrained MDP chases $\mathbb{E}[\sum_k \gamma^k r_k]$ and nothing else—no explicit budget on stalls or smoothness. So when the base policy is learned, we may recast streaming as a constrained MDP (CMDP) that maximizes expected fidelity under budgets on trajectory-level auxiliary costs [36]. Set $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ and $c_k^{\text{sm}} \equiv \text{smooth}_k$. The undiscounted episodic rebuffering ratio and mean smoothness cost read

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}, \quad (\text{A.1})$$

with targets $\delta_{\text{rf}}=0.05$ and $\delta_{\text{sm}}=3.5$ [3]. Per-step normalized costs $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ yield the Lagrangian-relaxed PPO reward

$$r_k^{\text{train}} = \frac{1}{100} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_t \psi_k^{\text{Lya}}), \quad (\text{A.2})$$

where projected ascent on the empirical constraint gaps updates the dual variables (λ_r, λ_s) [37]. Training may also subtract buffer-deviation and Lyapunov drift penalties [41]; these steady the optimization but certify nothing. The gap is real: finite-sample PPO does not certify per-chunk feasibility [5, 38], and CPS hands over that layer at deployment time.

The implementation rides Stable-Baselines 3 PPO [5, 43]: two Tanh MLPs (width 256), GAE ($\gamma=0.99$, $\lambda_{\text{GAE}}=0.95$), dual

step sizes $\eta_r=0.012$ and $\eta_s=0.003$, and projected ranges $\lambda_r \in [4.3, 40]$, $\lambda_s \in [0.3, 2.5]$ after a 60-episode warm-up. As in the main text, co-design wraps the training environment in CPS; shield-at-eval-only policies train without it.

Data availability

The CPS implementation, chunk-level emulator, evaluation scripts, per-title session-mean VMAF ladders, and per-episode result exports underlying Table 5–Table 11, Table 6, Table 7, and Fig. 3–Fig. 10 are available from the corresponding author upon reasonable request. A persistent archive with a DOI is planned for deposit upon acceptance. The FCC Measuring Broadband America and Norway/Nornet throughput traces used in the broadband suite are third-party public datasets available from their original providers [16, 17]. Synthetic 5G trace generators and configuration files used for the primary suite are included with the code release described above.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Leili Farzinvash: Conceptualization, Supervision, Project administration, Writing – review & editing.

Pedram Salehpour: Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude in order to improve language, clarity, and academic wording. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] T.-Y. Huang, R. Johari, N. McKeown, M. Trunnell, M. Watson, A buffer-based approach to rate adaptation: Evidence from a large video streaming service, in: *Proc. ACM SIGCOMM*, 2014, pp. 187–198.
- [2] K. Spiteri, R. Ugaonkar, R. K. Sitaraman, BOLA: Near-optimal bitrate adaptation for online videos, *IEEE/ACM Trans. Netw.* 28 (4) (2020) 1698–1711.
- [3] X. Yin, A. Jindal, V. Sekar, B. Sinopoli, A control-theoretic approach for dynamic adaptive video streaming over HTTP, in: *Proc. ACM SIGCOMM*, 2015, pp. 325–338.
- [4] H. Mao, M. Alizadeh, I. Menache, S. Kandula, Neural adaptive video streaming with pensieve, in: *Proc. ACM SIGCOMM*, 2017, pp. 197–210.
- [5] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, *arXiv:1707.06347* (2017).
- [6] A. Narayanan, X. Zhang, R. Zhu, A. Hassan, S. Jin, X. Zhu, X. Zhang, D. Rybkin, Z. Yang, Z. M. Mao, F. Qian, Z.-L. Zhang, A variegated look at 5G in the wild: Performance, power, and QoE implications, in: *Proc. ACM SIGCOMM*, 2021, pp. 610–625. doi:10.1145/3452296.3472923.
- [7] Cisco, Cisco annual internet report (2018–2023), <https://www.cisco.com/c/en/us/solutions/executive-perspectives/annual-internet-report/index.html>, accessed 2025 (2023).
- [8] M. Seufert, S. Egger, M. Slanina, T. Zinner, T. Hoffeld, P. Tran-Gia, A survey on quality of experience of HTTP adaptive streaming, *IEEE Commun. Surv. Tutor.* 17 (1) (2015) 469–492. doi:10.1109/COMST.2014.2360940.
- [9] Z. Li, A. Aaron, I. Katsavounidis, A. Moorthy, M. Manohara, Toward a practical perceptual video quality metric, *Netflix Technology Blog*, <https://netflixtechblog.com/toward-a-practical-perceptual-video-quality-metric-653f208b9652> (2016).
- [10] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, U. Topcu, Safe reinforcement learning via shielding, in: *Proc. AAAI Conf. Artificial Intelligence*, Vol. 32, 2018, pp. 2669–2678.
- [11] B. Könighofer, R. Bloem, N. Jansen, S. Junges, S. Pranger, Shields for safe reinforcement learning, *Commun. ACM* 68 (11) (2025) 80–90. doi:10.1145/3715958.
- [12] S. Gu, L. Yang, Y. Du, G. Chen, F. Walter, J. Wang, A. Knoll, A review of safe reinforcement learning: Methods, theories, and applications, *IEEE Trans. Pattern Anal. Mach. Intell.* 46 (12) (2024) 11216–11235. doi:10.1109/TPAMI.2024.3457538.
- [13] V. Vovk, A. Gammerman, G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005.
- [14] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, *Found. Trends Mach. Learn.* 16 (4) (2023) 494–591. doi:10.1561/2200000101.
- [15] D. Lakens, Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses, *Soc. Psychol. Personal. Sci.* 8 (4) (2017) 355–362. doi:10.1177/1948550617697177.
- [16] Federal Communications Commission, Measuring broadband america: Fixed broadband reports, <https://www.fcc.gov/reports-research/reports/measuring-broadband-america> (2015).
- [17] A. Kvalbein, D. Baltrūnas, K. Evensen, J. Xiang, A. Elmokashfi, S. Ferlin, The nornet edge platform for mobile broadband measurements, *Comput. Netw.* 61 (2014) 88–101.
- [18] Y. Sun, X. Yin, J. Jiang, V. Sekar, F. Lin, N. Wang, T. Liu, B. Sinopoli, CS2P: Improving video bitrate selection and adaptation with data-driven throughput prediction, in: *Proc. ACM SIGCOMM*, 2016, pp. 272–285.
- [19] A. Bentaleb, A. Taani, A. C. Begen, R. Zimmermann, S. Harous, A survey on bitrate adaptation schemes for streaming media over HTTP, *IEEE Commun. Surv. Tutor.* 21 (1) (2018) 562–585.
- [20] L. Peroni, S. Gorinsky, An end-to-end pipeline perspective on video streaming in best-effort networks: A survey and tutorial, *ACM Comput. Surv.* 57 (12) (2025). doi:10.1145/3742472.
- [21] R. A. K. Fezeu, E. Ramadan, W. Ye, B. Minneci, J. Xie, A. Narayanan, Z.-L. Zhang, et al., Unveiling the 5G mid-band landscape: From network deployment to performance and application QoE, in: *Proc. ACM SIGCOMM*, 2024. doi:10.1145/3651890.3672269.
- [22] I. Sodagar, The MPEG-DASH standard for multimedia streaming over the internet, *IEEE Multimed.* 18 (4) (2011) 62–67. doi:10.1109/MMUL.2011.71.
- [23] T. Chen, Y. Lin, N. Christianson, Z. Akhtar, S. Dharmaji, M. Hajiesmaili, A. Wierman, R. K. Sitaraman, SODA: An adaptive bitrate controller for consistent high-quality video streaming, in: *Proc. ACM SIGCOMM*, 2024. doi:10.1145/3651890.3672260.
- [24] A. Mehrabi, M. Siekkinen, A. Ylä-Jääski, Edge computing assisted adaptive mobile video streaming, *IEEE Trans. Mob. Comput.* 18 (4) (2019) 787–800. doi:10.1109/TMC.2018.2850026.
- [25] Z. Akhtar, Y. S. Nam, R. Govindan, S. Rao, J. Chen, E. Katz-Bassett, B. Ribeiro, Oboe: Auto-tuning video ABR algorithms to network conditions, in: *Proc. ACM SIGCOMM*, 2018, pp. 44–58.

- [26] M. Darwich, K. Khalil, Y. Ismail, A. Abdelgawad, M. Bayoumi, Conformal-ABR: Distribution-free risk-controlled bitrate selection for HTTP adaptive streaming, in: Proc. IEEE International Conference on Computing and Machine Intelligence (ICMI), 2026, pp. 1–6. doi:10.1109/ICMI68585.2026.11539758.
- [27] H. Xie, J. Zhu, Z. Shao, C. Fan, Z. Zhang, G. Yang, P. Luo, SafeSABR: Risk-calibrated adaptive bitrate streaming over Starlink networks (2026). arXiv:2605.23560, doi:10.48550/arXiv.2605.23560.
- [28] Z. Zhong, M. Liu, L. Yang, Y. Wang, Y. Xu, J.-N. Hwang, Video streaming with Kairos: An MPC-based ABR with streaming-aware throughput prediction, in: Proc. ACM Multimedia Systems Conference (MMSys), 2025, pp. 84–90. doi:10.1145/3712678.3721886.
- [29] M. Gadaleta, F. Chiariotti, M. Rossi, A. Zanella, D-DASH: A deep Q-learning framework for DASH video streaming, IEEE Trans. Cogn. Commun. Netw. 3 (4) (2017) 703–718. doi:10.1109/TCCN.2017.2755007.
- [30] M. Naresh, P. Saxena, M. Gupta, PPO-ABR: Proximal policy optimization based deep reinforcement learning for adaptive bitrate streaming, arXiv:2305.08114 (2023).
- [31] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2018.
- [32] T. Huang, C. Zhou, R.-X. Zhang, C. Wu, X. Yao, L. Sun, Comyco: Quality-aware adaptive video streaming via imitation learning, in: Proc. ACM International Conference on Multimedia (MM), 2019, pp. 429–437. doi:10.1145/3343031.3351014.
- [33] H. Zhang, A. Zhou, Y. Hu, C. Li, G. Wang, X. Zhang, H. Ma, L. Wu, A. Chen, C. Wu, Loki: Improving long tail performance of learning-based real-time video adaptation by fusing rule-based models, in: Proc. ACM MobiCom, 2021, pp. 775–788. doi:10.1145/3447993.3483259.
- [34] F. Y. Yan, H. Ayers, C. Zhu, S. Fouladi, J. Hong, K. Zhang, P. Levis, K. Winstein, Learning in situ: A randomized experiment in video streaming, in: Proc. USENIX NSDI, 2020, pp. 495–511.
- [35] X. Hu, A. Ghosh, X. Liu, Z.-L. Zhang, N. Shroff, COREL: Constrained reinforcement learning for video streaming ABR algorithm design over mmWave 5G, in: Proc. IEEE International Workshop on Communications Quality and Reliability (CQR), 2023, pp. 1–6. doi:10.1109/CQR59928.2023.10317803.
- [36] E. Altman, Constrained Markov Decision Processes, CRC Press, 1999.
- [37] S. Paternain, L. F. O. Chamon, M. Calvo-Fullana, A. Ribeiro, Constrained reinforcement learning has zero duality gap, in: Advances in Neural Information Processing Systems (NeurIPS), 2019.
- [38] J. Achiam, D. Held, A. Tamar, P. Abbeel, Constrained policy optimization, in: Proc. International Conf. Machine Learning (ICML), 2017, pp. 22–31.
- [39] Y. Chow, O. Nachum, E. Duenez-Guzman, M. Ghavamzadeh, A Lyapunov-based approach to safe reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2018.
- [40] V. V. Menon, H. Amirpour, M. Ghanbari, C. Timmerer, Perceptually-aware per-title encoding for adaptive video streaming, in: Proc. IEEE International Conference on Multimedia and Expo (ICME), 2022, pp. 1–6. doi:10.1109/ICME52920.2022.9859744.
- [41] M. J. Neely, Stochastic Network Optimization with Application to Communication and Queueing Systems, Morgan & Claypool, 2010.
- [42] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani, Conformal prediction beyond exchangeability, Ann. Stat. 51 (2) (2023) 816–845. doi:10.1214/23-AOS2276.
- [43] A. Raffin, A. Hill, M. Ernestus, A. Gleave, A. Kanervisto, N. Dormann, Stable-Baselines3: Reliable reinforcement learning implementations, J. Mach. Learn. Res. 22 (268) (2021) 1–8.
- [44] K. Spiteri, R. K. Sitaraman, D. Sparacio, From theory to practice: Improving bitrate adaptation in the DASH reference player, in: Proc. ACM Multimedia Systems Conference (MMSys), 2018, pp. 123–137. doi:10.1145/3204949.3204953.
- [45] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, J. Risk 2 (2000) 21–42.